The Architect of Oxygen-Sensing: A Comprehensive Analysis of William G. Kaelin Jr.'s Contributions to Molecular Oncology and Medicine

The evolution of modern cancer research is defined by the transition from descriptive pathology to the elucidation of fundamental biochemical circuits that govern cellular behavior. At the epicenter of this transformation is William G. Kaelin Jr., MD, a physician-scientist whose work on the von Hippel-Lindau (VHL) tumor suppressor gene fundamentally altered the understanding of how cells sense and adapt to oxygen availability.[1] Kaelin's career, culminating in the 2019 Nobel Prize in Physiology or Medicine, exemplifies a "disease-oriented" approach to basic science, where the investigation of rare genetic syndromes provides the "Rosetta Stone" for universal biological processes.[3] This report examines Kaelin's biographical trajectory, his landmark discoveries in the pVHL/HIF axis, the clinical translation of these findings into standard-of-care therapies for renal cell carcinoma and anemia, and his recent 2025 research into the intersections of hypoxia and tumor immunogenicity.

Clinical Foundations and the Evolution of a Scientist

The professional identity of William G. Kaelin Jr. is inextricably linked to his rigorous clinical training, which provided the deductive framework for his subsequent laboratory breakthroughs. Born on November 23, 1957, in New York City, Kaelin's early academic pursuits at Duke University were characterized by a focus on chemistry and mathematics, earning him a Bachelor of Science in both disciplines by 1979.[5] This quantitative background fostered a commitment to logical rigor and precise experimental control that would later define his laboratory's methodology.[7] Despite an early setback—a C- grade in undergraduate independent research where his professor suggested his future lay outside the laboratory—Kaelin pursued a medical degree at Duke University School of Medicine, graduating in 1982.[1]

Kaelin's clinical maturity was forged at the Johns Hopkins Hospital, where he served as a house officer and eventually as chief medical resident.[1] This period was critical, as it provided him with an encyclopedic understanding of human disease and the ability to recognize subtle clinical signals that pointed toward deeper biological truths.[3] In 1987, Kaelin relocated to Boston to begin a clinical fellowship in medical oncology at the Dana-Farber Cancer Institute (DFCI) and a postdoctoral fellowship in the laboratory of Dr. David Livingston.[1] Under Livingston's mentorship, Kaelin began his studies of tumor suppressor proteins, specifically the retinoblastoma (RB) protein, which regulates cellular replication.[1] This experience convinced Kaelin that the cancer genome itself could serve as a guide: by identifying recurrent mutations,

researchers could "let the cancer tell you which genes it cared about".[7] In 1992, Kaelin established his independent laboratory at DFCI, setting the stage for his investigation into the von Hippel-Lindau (VHL) gene.[5]

Career Phase	Institution	Tenure	Key Focus Areas
Undergraduate	Duke University	1975–1979	Chemistry, Mathematics [5]
Medical School	Duke University	1979–1982	Clinical Medicine [6]
Residency	Johns Hopkins Hospital	1982–1987	Internal Medicine, Chief Residency [1]
Oncology Fellowship	Dana-Farber Cancer Institute	1987–1991	Medical Oncology, RB Protein [1]
Independent Research	Dana-Farber Cancer Institute	1992–Present	VHL, Oxygen Sensing, HIF [5]
Academic Leadership	Harvard Medical School	2002–Present	Sidney Farber Professor of Medicine [1]
Institutional Governance	HHMI	1998–Present	Investigator [1]

The VHL Paradox: From Rare Syndrome to Biological Sentinel

The investigation of von Hippel-Lindau disease was a calculated risk that yielded a universal discovery. VHL syndrome is a rare, hereditary condition predisposing individuals to a variety of tumors, most notably clear cell renal cell carcinoma (ccRCC), central nervous system hemangioblastomas, and pheochromocytomas.[3] Kaelin's clinical observation was that VHL-associated tumors are remarkably vascularized—often described as "blood-vessel tumors"—and sometimes stimulate the overproduction of red blood cells (erythrocytosis).[3] These traits are the body's physiological responses to low oxygen, yet they occur in VHL patients even when oxygen is abundant.[3]

In 1996, Kaelin's laboratory published findings demonstrating that cells lacking a functional VHL

gene were incapable of sensing oxygen.[3] These cells continuously expressed hypoxia-inducible genes, such as Vascular Endothelial Growth Factor (VEGF) and Erythropoietin (EPO), regardless of the actual oxygen concentration in the environment.[3] When a functional VHL gene was reintroduced into these cells, the normal oxygen-sensing response was restored.[17] This established the pVHL protein (the product of the VHL gene) as a central regulator of the hypoxia response pathway. The focus then shifted to the biochemical mechanism by which pVHL performed this function, eventually converging with the research of Gregg Semenza and Peter Ratcliffe.[17]

The Convergence of pVHL and Hypoxia-Inducible Factors

While Kaelin approached the problem through the lens of tumor suppression, Gregg Semenza at Johns Hopkins was investigating how the EPO gene is regulated in response to low oxygen.[17] Semenza identified a two-part protein complex he named Hypoxia-Inducible Factor (HIF), consisting of an oxygen-labile α-subunit ($HIF-1\alpha$) and a stable β-subunit.[18] Simultaneously, Peter Ratcliffe at Oxford demonstrated that this oxygen-sensing mechanism was present in virtually all mammalian cell types, not just those in the kidney or liver.[17]

The critical link was the discovery that pVHL serves as the substrate-recognition component of an E3 ubiquitin ligase complex.[13] In the late 1990s, the field realized that in the presence of oxygen, pVHL targets $HIF\alpha$ for polyubiquitination and subsequent degradation by the 26S proteasome.[3] This molecular "garbage disposal" system ensures that HIF levels remain low when oxygen is plentiful.[3] When oxygen is absent or the VHL gene is mutated, HIF is stabilized, enters the nucleus, and activates hundreds of genes that promote survival in low-oxygen conditions.[13]

The Biochemical Switch: Prolyl Hydroxylation

The crowning achievement of Kaelin's mechanistic research came in 2001, when he and Ratcliffe independently elucidated the exact nature of the oxygen signal.[17] They discovered that the interaction between pVHL and $HIF\alpha$ is governed by a specific post-translational modification: prolyl hydroxylation.[6] This chemical modification involves the addition of a hydroxyl group (-OH) to specific proline residues (specifically Pro564 and Pro402 in human $HIF-1\alpha$).[18]

The Role of Prolyl Hydroxylases (PHDs)

The enzymes responsible for this modification, known as Prolyl Hydroxylase Domain (PHD) proteins or EglN proteins, act as the actual oxygen sensors of the cell.[13] These enzymes require molecular oxygen (O_2), iron (Fe^{2+}), and 2-oxoglutarate to function.[13] Under normoxic

conditions, the PHDs are active and hydroxylate $HIF\alpha$, creating a binding site for the pVHL protein.[18] Without this hydroxylation, pVHL cannot recognize $HIF\alpha$.[18] This elegant biochemical switch explains how a physical change in the environment (oxygen concentration) is directly transduced into a biological signal (protein stability and gene expression).

Kaelin's 2001 paper utilized precise experimental models to confirm this, including human cell systems capable of hydroxylation and plant cell systems that were not, showing that pVHL only recognized $HIF\alpha$ when the appropriate chemical modification was present.[20] This discovery was fundamental because it defined the molecular target for new drugs aimed at treating anemia and cancer.[15]

Component of the pVHL/HIF Axis	**Function/Role**	**Dependency**
HIF-α Subunit	Transcription factor that activates hypoxia-response genes [18]	Degraded in high O_2; stable in low O_2
pVHL Protein	Ubiquitin ligase that targets hydroxylated HIF-α [13]	Requires HIF-α prolyl hydroxylation
Prolyl Hydroxylase (PHD)	Enzyme that adds hydroxyl groups to HIF-α proline residues [20]	Requires O_2, Fe^{2+}, 2-oxoglutarate
Ubiquitin Ligase Complex	Multi-protein assembly (VHL, Elongin B/C, Cul2, Rbx1) [18]	Structural scaffold for HIF degradation
Hypoxia Response Element (HRE)	DNA binding site for the stabilized HIF complex [21]	Requires HIF-α/HIF-β dimerization

Therapeutic Revolutions in Renal Cell Carcinoma

Kaelin's research had immediate and profound implications for the treatment of clear cell renal cell carcinoma (ccRCC), the most common form of kidney cancer.[14] Because ccRCC is characterized by the loss of the VHL gene, these tumors are driven by the inappropriate stabilization of HIF and the overproduction of VEGF.[13] This provided a clear therapeutic target:

the VEGF pathway.

The Success of VEGF Inhibitors

The molecular rationale provided by Kaelin helped motivate the clinical testing of VEGF inhibitors.[9] Drugs such as sunitinib and sorafenib, which block the receptors for VEGF, became the standard of care for advanced RCC between 2005 and 2015, dramatically improving survival rates.[14] However, Kaelin recognized that while VEGF inhibitors were a major leap forward, they were rarely curative and often faced the development of resistance.[21]

Drugging the "Undruggable" HIF-2α

Kaelin's laboratory went further to identify which isoform of HIF was the primary driver of kidney cancer. While both $HIF-1\alpha$ and $HIF-2\alpha$ are present, Kaelin's group demonstrated that $HIF-2\alpha$ is the key oncogenic driver in ccRCC.[4] For years, transcription factors like HIF were considered "undruggable" because they lacked traditional ligand-binding pockets.[8] Kaelin challenged this "truism" and supported the development of small-molecule inhibitors that could bind to a specific pocket in $HIF-2\alpha$ and prevent its activity.[8]

This effort led to the development and 2021 FDA approval of belzutifan (Welireg), the first $HIF-2\alpha$ inhibitor.[1] Originally approved for VHL-associated tumors, it is now being evaluated in sporadic kidney cancer and other malignancies.[14] This transition from systemic anti-angiogenesis (VEGF inhibition) to selective transcription factor inhibition (HIF-2 inhibition) represents a move toward greater specificity and reduced toxicity in oncology.[14]

The Global Management of Anemia via HIF-PH Inhibitors

The same oxygen-sensing mechanism that drives cancer also regulates the production of red blood cells. In patients with chronic kidney disease (CKD), the body fails to produce enough EPO, leading to chronic anemia.[3] Kaelin realized that by inhibiting the PHD enzymes, one could "trick" the body into thinking it was hypoxic, thereby stabilizing HIF and inducing the production of endogenous EPO in the liver or remaining kidney tissue.[3]

This concept birthed a new class of oral drugs: HIF-Prolyl Hydroxylase Inhibitors (HIF-PHIs). These medications offer a more physiological approach to treating anemia than standard injectable erythropoiesis-stimulating agents (ESAs).[23]

Regulatory Status and the Roxadustat Dilemma

The global rollout of HIF-PHIs has been uneven. Roxadustat, developed by FibroGen (where

Kaelin has served as a scientific advisor), was the first-in-class drug approved in China in 2018.[25] However, its path in the United States was derailed in 2021 when the FDA rejected its marketing application due to safety concerns and data transparency issues.[25] Despite this, other HIF-PHIs have successfully navigated the U.S. regulatory landscape for dialysis patients:

Drug Molecule	Brand Name	Mechanism	Primary Indication	FDA Approval Date
Daprodustat	Jesduvroq	HIF-PH Inhibitor	Anemia of CKD (on dialysis) [26]	February 1, 2023
Vadadustat	Vafseo	HIF-PH Inhibitor	Anemia of CKD (on dialysis) [23]	March 28, 2024
Belzutifan	Welireg	HIF-2α Inhibitor	VHL-associated Tumors / RCC [22]	August 13, 2021

These approvals represent the clinical culmination of three decades of basic research into the VHL-HIF axis, providing oral treatment options that potentially reduce the cardiovascular risks associated with high-dose ESA therapy.[24]

Frontiers of 2025: Endogenous Retroviruses and Immunotherapy

Even after receiving the Nobel Prize, Kaelin's laboratory continues to push the boundaries of cancer biology. In early 2025, Kaelin co-authored a landmark study in *Cell* that linked HIF regulation to the activation of latent viral sequences within the human genome.[1]

The Mechanism of Viral Mimicry in Kidney Cancer

Clear cell renal cell carcinoma is uniquely immunogenic despite having a low mutational burden.[28] The 2025 research titled "HIF regulates multiple translated endogenous retroviruses: Implications for cancer immunotherapy" provides a novel explanation for this phenomenon.[29]

The study found that $HIF-2\alpha$, which is overabundant in VHL-deficient tumors, drives the reactivation of endogenous retroviruses (ERVs)—ancient viral DNA integrated into our genome millions of years ago that is normally silenced.[28]

The implications of this finding are transformative:

- **Antigen Presentation:** These reactivated ERVs are translated into proteins, which the cancer cells then process into peptide fragments.[28]

- **Immune Flagging:** These fragments are presented on the cell surface as HLA-bound "flags" that the immune system, specifically T cells, can recognize as foreign.[28]
- **Therapeutic Induction:** The research showed that ERV expression can be induced in non-VHL-mutant tumors using clinical-grade HIF stabilizers, potentially turning "cold" tumors "hot" for immunotherapy.[29]

This discovery points toward a future where "viral mimicry" can be therapeutically exploited to enhance the efficacy of checkpoint inhibitors and cancer vaccines, not only in kidney cancer but across a broader spectrum of malignancies.[28]

Scientific Philosophy and Advocacy for Discovery

Kaelin is a staunch defender of the "Science Golden Goose"—curiosity-driven basic research.[4] In his 2025 commentary in the *American Journal of Medicine*, Kaelin argues that the modern emphasis on immediate "translational" outcomes risks stifling the very foundational discoveries that make translation possible.[1] He often shares the story of his C- grade in undergraduate research to encourage young scientists that "a prize is a tribute to nature, not a tribute to us".[4]

The "Combination Lock" Analogy

In his 2024 keynote address at the San Antonio Breast Cancer Symposium, Kaelin introduced a refined model for cancer drug development based on the "combination lock" analogy.[22] He posited that while multiple mutations (tumblers) are necessary to initiate cancer, correcting or targeting even one of the *early* clonal mutations is often sufficient to selectively kill the tumor cells.[22] This logic drives his current focus on targeting early-stage genetic lesions (like VHL, RB, or IDH) to overcome the problem of intratumoral heterogeneity.[11]

Rigor and Reproducibility

Kaelin is also a vocal critic of "sloppy" preclinical research. He implores scientists to use robust positive and negative controls and to actively seek the *least* interesting explanation for their data before drawing conclusions.[7] This commitment to scientific discipline is, in his view, more important than learning any specific technology, as technologies go out of style while clear, logical thinking does not.[7]

Institutional Leadership and Governance

Kaelin's influence extends beyond the bench through his roles in institutional governance and corporate advisory. As a Sidney Farber Professor of Medicine at DFCI and Harvard, he oversees a major research enterprise while serving on the Board of Directors for Eli Lilly and Company and, as of November 2025, the global clinical services provider IQVIA.[5] These roles allow him to bridge the gap between discovery and the commercialization of new therapies, ensuring that mechanistic insights are translated into clinical practice.[10]

Furthermore, he remains active in the scientific community through his service on the National

Cancer Institute (NCI) Board of Scientific Advisors and the Damon Runyon Cancer Research Foundation.[5] His recent awards, such as the Stanley P. Reimann Honor Award (2024) and the NCI Outstanding Investigator Award (2023), underscore his ongoing relevance as a leader in molecular oncology.[34]

Notable Award	Year	Significance
Nobel Prize in Physiology or Medicine	2019	Recognition of oxygen-sensing discovery [2]
Albert Lasker Basic Medical Research Award	2016	America's most prestigious biomedical honor [1]
Canada Gairdner International Award	2010	Recognition of transformative medical research [1]
Stanley P. Reimann Honor Award	2024	Fox Chase Cancer Center's highest distinction [36]
ASCO Science of Oncology Award	2016	Impact on clinical oncology and drug development [1]
Duke Distinguished Alumni Award	2007	Recognition from his medical alma mater [1]
NCI Outstanding Investigator Award	2023	Support for continued excellence in research [35]

Conclusion: The Legacy of Mechanistic Inquiry

The contributions of William G. Kaelin Jr. have fundamentally redefined the boundaries of both biology and medicine. By relentlessly pursuing the mechanism behind a rare genetic disorder, he uncovered a universal system that allows all multicellular life to survive in an oxygenated atmosphere.[4] This discovery has not only saved lives through the development of VEGF and HIF-2 inhibitors for kidney cancer but has also provided hope for millions of patients with chronic anemia.[10]

As Kaelin enters the 2025-2026 research cycle, his focus remains on "drugging the undruggable" and leveraging ancient viral sequences to harness the power of the immune

system.[11] His work serves as a reminder that the most significant medical breakthroughs often begin with a "light bulb moment" rooted in basic, curiosity-driven exploration.[7] Through his commitment to scientific rigor, logical clarity, and institutional leadership, William Kaelin Jr. continues to architect the future of precision medicine, ensuring that the "Science Golden Goose" continues to thrive for the benefit of humankind.[2]

Works cited

1. William G. Kaelin Jr., MD - Dana-Farber Cancer Institute | Boston, MA, accessed on January 22, 2026, https://www.dana-farber.org/find-a-doctor/william-g-kaelin-jr
2. William G. Kaelin Jr – Facts – 2019 - NobelPrize.org, accessed on January 22, 2026, https://www.nobelprize.org/prizes/medicine/2019/kaelin/facts/
3. From a Rare Cancer Gene to a Fundamental Life Process: The Nobel Prize-Winning Research of William G. Kaelin, Jr, MD, accessed on January 22, 2026, https://blog.dana-farber.org/insight/2019/10/from-a-rare-cancer-gene-to-a-fundamental-life-process-the-nobel-prize-winning-research-of-william-g-kaelin-jr-md/
4. William Kaelin's 2019 Nobel Prize in Physiology or Medicine - Dana-Farber Cancer Institute, accessed on January 22, 2026, https://www.dana-farber.org/about/awards/william-kaelin-2019-nobel-prize
5. William Kaelin Jr. - Wikipedia, accessed on January 22, 2026, https://en.wikipedia.org/wiki/William_Kaelin_Jr.
6. William G. Kaelin, Jr. | Biography, Facts, & Nobel Prize | Britannica, accessed on January 22, 2026, https://www.britannica.com/biography/William-G-Kaelin-Jr
7. Dr. Bill Kaelin Studies Tumor Suppressors to Improve the Understanding of Cancer Biology, accessed on January 22, 2026, https://www.cancer.gov/about-nci/organization/dcb/progress/meet-investigators/bill-kaelin
8. Nobel laureate William Kaelin draws a crowd | Cold Spring Harbor Laboratory, accessed on January 22, 2026, https://www.cshl.edu/labdish/nobel-laureate-william-kaelin-draws-a-crowd/
9. William Kaelin - Kaelin Lab - Cancer Biology, accessed on January 22, 2026, https://kaelinlab.dana-farber.org/william-kaelin.html
10. Episode 1 | Interview with Nobel Prize winning physician-scientist William Kaelin, Jr., accessed on January 22, 2026, https://freseniusmedicalcare.com/en-us/insights/gmo-dialogs/episode-1--interview-with-nobel-prize-winning-physician-scientis/
11. William Kaelin | HMS Office for Graduate Education PhD Programs, accessed on January 22, 2026, https://ogephd.hms.harvard.edu/people/william-kaelin
12. William Kaelin Jr., MD | Harrington Discovery Institute at University Hospitals, accessed on January 22, 2026, https://www.harringtondiscovery.org/about/scientific-advisory-board/william-kaelin-jr-md

13. The VHL Tumor Suppressor Gene: Insights into Oxygen Sensing and Cancer - PMC, accessed on January 22, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5525432/
14. How 2019 Nobel-Winner Kaelin's Work Changed the Treatment of Kidney Cancer, accessed on January 22, 2026, https://www.brighamhealthonamission.org/2019/12/04/how-2019-nobel-winner-kaelins-work-changed-the-treatment-of-kidney-cancer/
15. Gene linked to kidney cancer led to a Nobel Prize, accessed on January 22, 2026, https://www.kidneycancer.org/gene-led-to-nobel-prize/
16. HMS Professor Kaelin wins Nobel in Physiology or Medicine | Harvard Medical School, accessed on January 22, 2026, https://hms.harvard.edu/news/hms-professor-kaelin-wins-nobel-physiology-or-medicine
17. The Nobel Prize in Physiology or Medicine 2019 - Press release - NobelPrize.org, accessed on January 22, 2026, https://www.nobelprize.org/prizes/medicine/2019/press-release/
18. Oxygen sensing and adaptability won the 2019 Nobel Prize in Physiology or medicine - NIH, accessed on January 22, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6889041/
19. Penn alumnus Gregg L. Semenza awarded Nobel in Physiology or Medicine 2019, accessed on January 22, 2026, https://penntoday.upenn.edu/news/penn-alumnus-gregg-l-semenza-awarded-nobel-physiology-or-medicine-2019
20. Analyzing a Nobel-winning paper with AI - Bay Bridge Bio, accessed on January 22, 2026, https://www.baybridgebio.com/blog/kaelin-2001
21. William G. Kaelin Jr. and Gregg L. Semenza receive the 2012 ASCI/Stanley J. Korsmeyer Award - NIH, accessed on January 22, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3314483/
22. Keynote speaker: Questioning assumptions, ignoring naysayers are ..., accessed on January 22, 2026, https://www.sabcsmeetingnews.org/keynote-speaker-questioning-assumptions-ignoring-naysayers-are-key-to-drug-development/
23. DRUG NAME Vafseo (vadadustat) - CareSource, accessed on January 22, 2026, https://www.caresource.com/documents/medicaid-in-policy-pharmacy-vafseo-20251001
24. 215192Orig1s000 OTHER REVIEW(S) - accessdata.fda.gov, accessed on January 22, 2026, https://www.accessdata.fda.gov/drugsatfda_docs/nda/2024/215192Orig1s000OtherR.pdf
25. The Past and Present of 'Miraculous Medicine' Roxadustat - 药时代DrugTimes, accessed on January 22, 2026, https://www.drugtimes.cn/2024/07/12/thepastandpresentofmiraculousmedicineroxadustat/
26. JESDUVROQ (daprodustat) tablets - This label may not be the latest approved by FDA. For current labeling information, please visit https://www.fda.gov/drugsatfda,

accessed on January 22, 2026, https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/216951s000lbl.pdf
27. Understanding the TDAPA for Treating Anemia in Patients on Dialysis in - Kidney News Online, accessed on January 22, 2026, https://www.kidneynews.org/view/journals/kidney-news/16/10/11/article-p9_3.xml
28. Latent viruses in human genome inspire novel directions for kidney ..., accessed on January 22, 2026, https://www.dana-farber.org/newsroom/news-releases/2025/latent-viruses-in-human-genome-inspire-novel-directions-for-kidney-cancer-immunotherapy
29. HIF regulates multiple translated endogenous retroviruses: Implications for cancer immunotherapy - PubMed, accessed on January 22, 2026, https://pubmed.ncbi.nlm.nih.gov/40023154/
30. HIF regulates multiple translated endogenous retroviruses: Implications for cancer immunotherapy. - National Genomics Data Center (CNCB-NGDC), accessed on January 22, 2026, https://ngdc.cncb.ac.cn/openlb/publication/OLB-PM-40023154
31. HIF Regulates Multiple Translated Endogenous Retroviruses: Implications for Cancer Immunotherapy | Request PDF - ResearchGate, accessed on January 22, 2026, https://www.researchgate.net/publication/389433663_HIF_regulates_multiple_translated_endogenous_retroviruses_Implications_for_cancer_immunotherapy
32. William Kaelin | Harvard Catalyst Profiles | Harvard Catalyst, accessed on January 22, 2026, https://connects.catalyst.harvard.edu/Profiles/profile/1249231
33. Dr. William G. Kaelin Jr. joins IQVIA Board of Directors - Investor Relations, accessed on January 22, 2026, https://ir.iqvia.com/press-releases/press-release-details/2025/Dr--William-G--Kaelin-Jr--joins-IQVIA-Board-of-Directors/default.aspx
34. Noteworthy | American Academy of Arts and Sciences, accessed on January 22, 2026, https://www.amacad.org/bulletin/spring-2024/noteworthy
35. Dana-Farber researchers receive prestigious 2023 Outstanding Investigator Award from National Cancer Institute, accessed on January 22, 2026, https://www.dana-farber.org/newsroom/news-releases/2024/dana-farber-researchers-receive-prestigious-2023-outstanding-investigator-award-from-national-cancer-institute
36. Fox Chase Cancer Center Recognizes Nobel Laureate Dr. William Kaelin With the Stanley P. Reimann Honor Award, accessed on January 22, 2026, https://www.foxchase.org/news/2024-04-04-fox-chase-cancer-center-recognizes-nobel-laureate-dr-william-kaelin-with-the-stanley-p-reimann-honor-award